BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Timed Paper 1 Practice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the purpose of the Data Protection Act 2018?

2. Why indent code?

A) To fix every logic error
B) To replace all conditions
C) To change false into true
D) To show blocks clearly

3. "A comment always changes the value of the variable on the next line." Correct this. Explain why.

4. An indicator uses Q = NOT A. Give outputs for A=0 and A=1, then evaluate NOT(NOT A) for both values.

Today's objectives

Explain the core ideas of timed paper 1 practice using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Paper 1 timing
The OCR J277/01 paper has 80 marks in 90 minutes.

Show calculations
Write formula, substitute numbers and give final units.

Plan explanations
Match distinct points and justifications to mark tariff.

Track time
Leave opportunity to revisit unanswered questions without sacrificing required detail.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Paper 1 timing	The OCR J277/01 paper has 80 marks in 90 minutes.
Show calculations	Write formula, substitute numbers and give final units.
Plan explanations	Match distinct points and justifications to mark tariff.
Track time	Leave opportunity to revisit unanswered questions without sacrificing required detail.

I Do - One worked example

Model a four-mark image-size calculation: 80 x 50 pixels at 8 bits/pixel.

1. Count 80 x 50=4000 pixels.
2. Multiply by eight: 32,000 bits.
3. Divide by eight: 4000 bytes.
4. Check that metadata/compression are excluded.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Plan a six-mark comparison of SSD and HDD for a traveller.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Convert a raw data size from bits to bytes.

```
sizeBits = 24000
sizeBytes = sizeBits DIV 8
print(sizeBytes)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State RAM volatility and its implication. [2 marks]

Q3. Calculate the largest eight-bit unsigned number. [2 marks]

Q4. Explain why a router is needed between separate networks. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Units can be ignored as long as the final calculation looks plausible. ☐ T ☐ F
2. The OCR J277/01 paper has 80 marks in 90 minutes. ☐ T ☐ F
3. Write formula, substitute numbers and give final units. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It sets rules for using personal data lawfully and protects people's rights. ☐

Q2 D - To show blocks clearly ☐

Q3 It explains important intent or reasoning without misleading the reader. ☐

Q4 (a) Q: 1, 0. Double NOT: 0, 1, restoring A. ☐

Code First - Model

24,000 bits / 8 = 3,000 bytes (3 kB using decimal units). Changing to 48,000 bits gives 6,000 bytes (6 kB). This is a unit calculation, not a complete file-size formula.

We Do - Check the model

1. Balance with higher cost per GB; link to travel needs. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Volatile; active contents are lost without power. ☐

Q3 255. ☐

Q4 Forwards packets using logical/IP routing, allowing communication between networks. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: